

Nitheesh Karthikeyan

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Engineering student with hands-on experience in software development, data analytics, and client-facing technology delivery - passionate about designing innovative, data-driven solutions to solve real-world business problems.

EDUCATION

VELLORE INSTITUTE OF TECHNOLOGY

2024--2028 | Chennai, India
B.Tech in Electronics & Communication Engineering

SBOA SCHOOL & JR. COLLEGE

2010--2024 | Chennai, India
Multiple times winner in **Model United Nations**; strong structured communication, negotiation & policy analysis.

SKILL SET

Programming:

Python, Java, C++, C, SQL

Data & Analytics:

Tableau, Google Analytics, Mixpanel, NumPy, Pandas, Scikit-Learn, Excel

PM & Agile Tools:

Jira, Notion, Figma, Agile/Scrum

Developer Tools:

Git, Docker, Next.js, React, $\LaTeX$

Product Analytics:

Funnel Analysis, A/B Testing, User Journey Mapping, DAU/MAU, CAC/LTV

LANGUAGES

English — Fluent Public Speaker

Tamil — Native

Hindi — Professional Working

French — Limited Proficiency

LEADERSHIP

UN Millennium Fellow 2025 (UNAI) — Top 4% of 60,000+ applicants; led sustainability-driven programs across 3+ communities; 10+ workshops reaching 200+ students.

Paadhai (Career Awareness) — Monthly sessions bridging career gaps for underserved govt. school students; 150+ impacted.

HackClub VIT Chennai (Jr. Core) — Led events with tech CEOs; coordinated cross-functional teams. **Best Club in VIT 2x.**

CERTIFICATIONS

Deloitte Australia (Forage) Jul 2025

Data analytics simulation; built Tableau dashboard and performed structured data analysis in Excel.

Siemens Mobility (Forage) Mar 2026

Commercial PM simulation; analyzed \$100M+ project using WBS, KPIs, and EAC; stakeholder reporting.

JPMorgan Chase (Forage) Jul 2025

Investment banking simulation; built DCF model and evaluated M&A target.

REFERENCES

Pranay V. Jain

Alumni, EY Chennai

Assurance Associate

(Oct 2021 – May 2022)

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EXPERIENCE

SAMSUNG RESEARCH INDIA - BENGALURU | Student Researcher

February 2026 -- September 2026 | Hybrid, Self-Paced

- Selected for **Samsung PRISM** — competitive industry-academia program; one of only **6 students** from VIT, working directly under two research scientists at SRI-B.
- Conducting applied research on **Multi-Core Tensorization of LLMs via Tensor Train Decomposition** — compressing 7B+ parameter models with <10% performance degradation.
- Executing a structured, phase-gated research roadmap: literature survey of LLM & Tensor Network state-of-the-art → comparative algorithm study (MPS, MPO, SVD) → baseline implementation → **large-scale HPC cluster optimization** → technical publication drafting.
- Applying sensitivity analysis & hyper-parameter tuning for optimal bond dimensions across multi-core TT decomposition configurations on high-performance computing infrastructure.

MEDXAI | Research Software Intern (Stipend)

October 2024 -- February 2026

- Primary **technology delivery partner** for **20+ pharmaceutical clients** — full cycle from requirement gathering through iterative development, testing, and deployment, mirroring a **consulting engagement lifecycle**.
- Engineered **Drug Explorer**: SMILES-to-insight data pipeline (binding affinity, drug synergy); now **outsourced to and actively used by major pharmaceutical companies**.
- Contributed to a drug candidate passing **wet-lab validation** from computational predictions — translating software outputs into measurable real-world health impact.
- Operated within **Agile/Scrum** environments across the full SDLC — sprint planning, daily stand-ups, and iterative release cycles using Jira; collaborated with cross-functional global teams.
- Produced data-oriented reports, dashboards, and client documentation for multinational clients including OmniActive, translating complex technical outputs into clear, executive-ready insights.
- Collaborated with interdisciplinary teams across research, engineering, and healthcare — building stakeholder communication and requirements translation skills essential for consulting contexts.

FREELANCE | Web Developer & UI/UX Designer

December 2025 -- Present

- Designed & developed user-centric digital products for **8+ clients** — full SDLC from discovery to deployment; improved usability & retention by iterating on behavioral analytics and data-driven design thinking.

ZERODHA VARSITY | Lead Coordinator - Chennai Edition

October 2025 -- February 2026

- Managed cross-functional teams for city-wide financial literacy competition: **500+ participants**, **INR 3.5L+ prize pool**; 3-month execution with rigorous stakeholder accountability; **re-invited for two subsequent external engagements** based on prior competence of execution.

PROJECTS AS PRODUCTS

- LithoMap — Clinical Research Data Viz Platform**: Production tool used daily at Govt. Stanley Medical College to encode & visualize FTIR clinical data; full design-to-deployment cycle with domain experts.
- ML-Based Library Seat Occupancy System**: Real-time CV & ML system tracking seat availability; actively deployed in VIT campus library. Awarded **VP's Feather-in-the-Cap Award, VIT**.
- Financial Fraud Detection** (Qinnovision 2025, NYU Abu Dhabi): Privacy-first anomaly detection via quantum federated ML for distributed financial data.
- Pothole Detection System**: Computer vision-based road anomaly detection system; recognized among Top 3 Non-Startup Projects at VIT Chennai 2024 (Iliventiv, IIT Madras). Real-world perception task analogous to ADAS object detection pipelines.

HACKATHONS

- IBM Qiskit Fall Fest 2024 — 1st Place**. Presented to senior leadership (Dr. L. Venkataraman) at IBM Quantum India; reviewed by international faculty.
- IBM Qiskit Fall Fest 2025 — Judge & Teaching Instructor**. Delivered sessions to **100+ participants** from 10+ countries; facilitated live mentoring.
- Intel + Microsoft SAS Hackathon (HEC Montreal) — World Rank 19**. Quantitative user-flow analysis to optimize business outcomes.
- Goldman Sachs Quantitative Finance Hackathon — AIR 163**. Alpha-generating strategies via portfolio analytics & quantitative modelling.

AWARDS

- UN Millennium Fellow (2025)** — Top 4% of 60,000+ applicants; selected for global academic & social impact through technology.
- 1st Place — Best Startup Idea (2025)** — Presented by Dr. Soumya Swaminathan, former Chief Scientist, WHO.
- Nokia Campus Connect** — 1 of 5 from VIT Chennai; training in communication systems, ML & CS.
- VP's Feather-in-the-Cap Award, VIT** — ML-based Seat Occupancy System deployed in VIT Library.